

Output Devices

- | | |
|-------------|-----------|
| ① Monitor | ③ Plotter |
| ② Printer | ④ Speaker |
| ⑤ Projector | |

Input

- | | |
|---------------|-------------------|
| 1. Barcode | 4. Digital Camera |
| 2. QR code | 5. Keyboard |
| 3. Microphone | 6. Mouse |
| 7. Scanner | 8. Touch Screen |

Input device is used to enter the data into the CPU.

Output devices are used to produce information/result from CPU.

Printer → is an output device used to produce result/output on paper

1. Inkjet (cartridge)
2. Laser (Toner)
3. 3D printer

Inkjet

✓ 1- Inexpensive to buy
operational cost
is high

✓ 2- low pages per
cartridge

✓ 3- Slow rate
of printing

✓ 4- Compact in

5- Paper ^{size} tray is small
Projector is used to

Laser

✓ 1- Expensive to
buy but operational
cost is low

✓ 2- Large volume
of printing

✓ 3- High / faster
rate of printing

✓ 4- Takes more

5- Has ^{space} larger
paper tray

larger output - on screen.

① Cinema

② Shopping Malls

3- School

4- University

LED → light-emitting diode

LCD → liquid crystal Display

Local speakers

1- Schools

2- Shopping Mall

3- Airport for flight
arrival

4- Super Market for
product promotions

Monitoring and Control

- ① Sensor
- ② ADC
- ③ Microprocessor (check pre-stored)
- ④ DAC
- ⑤ Actuator / Motor
- ⑥ Process is continuous

Memory

- Small in size
4 GB, 16, 32, 64
- Temporary
- Volatile

Storage

- Huge in size
- 4 TB
- Permanent
- Non-Volatile

CD	← optical Red laser	→ DVD
Single layer		Dual layer
700 MB		4.7 GB
780 nm (wavelength)		650 nm (wavelength)